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Merge Data pada Spoon

1. Teori Singkat

Data warehouse adalah jenis sistem manajemen data yang dirancang untuk memungkinkan dan mendukung kegiatan business intelligence (BI), terutama analitik. Gudang data semata-mata dimaksudkan untuk melakukan kueri dan analisis dan sering berisi sejumlah besar data historis. Data dalam gudang data biasanya berasal dari berbagai sumber seperti file log aplikasi dan aplikasi transaksi. Gudang data memusatkan dan mengkonsolidasikan sejumlah besar data dari berbagai sumber. Kemampuan analitisnya memungkinkan organisasi untuk memperoleh wawasan bisnis yang berharga dari data mereka untuk meningkatkan pengambilan keputusan. Seiring waktu, ia membangun catatan sejarah yang dapat sangat berharga bagi para ilmuwan data dan analisis bisnis. Karena kemampuan ini, gudang data dapat dianggap sebagai "sumber kebenaran tunggal" organisasi.

2. Alat dan Bahan

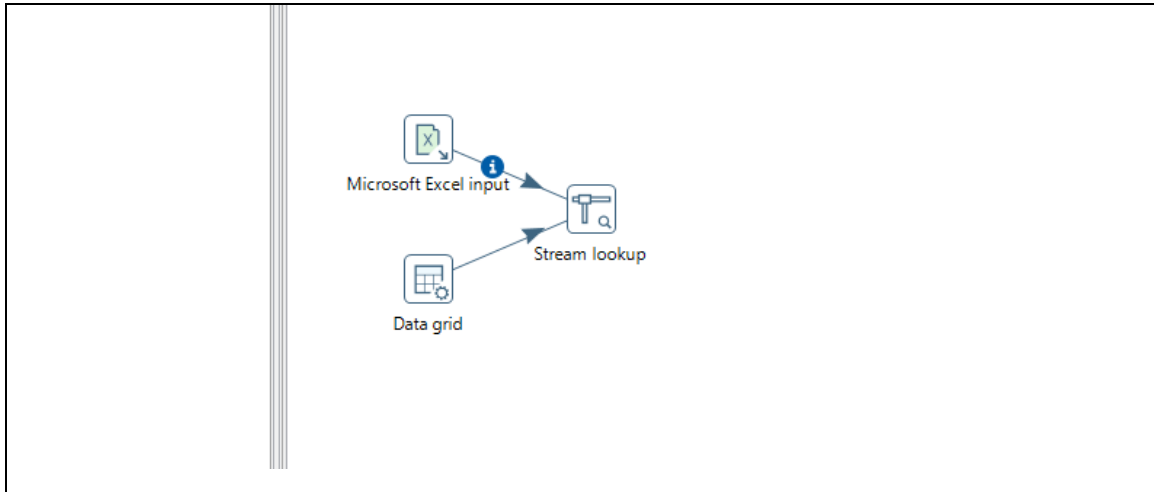
Hardware : Laptop/PC

Software : Spoon Pentaho from Hitachi Vantara

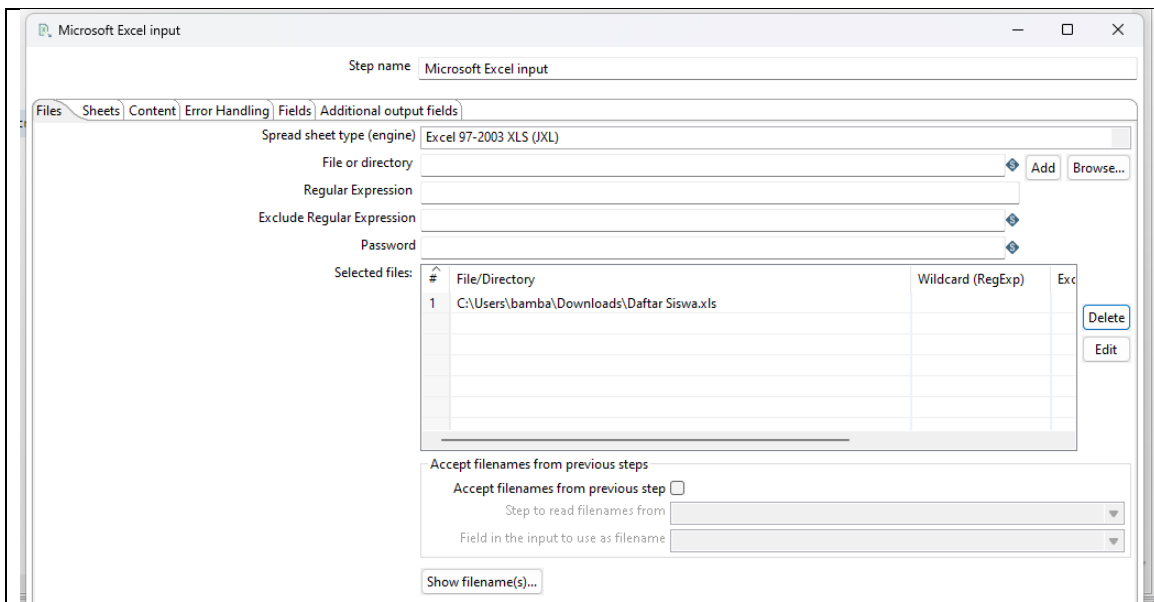


3. Elemen Kompetensi

- a. Latihan pertama – Melakukan penggabungan data menggunakan Stream Lookup
1. Buat transformation baru dengan struktur seperti pada gambar dibawah ini.



2. Microsoft Excel Input.



Microsoft Excel input

Step nameMicrosoft Excel Input

FilesSheetsContentError HandlingFieldsAdditional output fields

List of sheets to read

#	Sheet name	Start row	Start column
1	Sheet1	0	0

Get sheetname(s)...

HelpOKPreview rowsCancel

Microsoft Excel input

Step nameMicrosoft Excel input

FilesSheetsContentError HandlingFieldsAdditional output fields

#	Name	Type	Length	Precision	Trim type	Repeat	Format	Currency	Decimal	Grouping
1	NIS	String	-1	-1	none	N				
2	Nama	String	-1	-1	none	N				
3	Jenis Kelamin	String	-1	-1	none	N				
4	Kelas	String	-1	-1	none	N				

Get fields from header row...



3. Data Grid.

Step name: Data grid

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Null if	Set empty string?
1	Nama	String								N
2	Kelas	String								N

Buttons: Help, OK, Preview, Cancel

Step name: Data Grid

#	Nama	Kelas
1	MAWAR MELATI	A
2	MAWAR MELATI	B

Buttons: Help, OK, Preview, Cancel

4. Stream lookup (Get fields, Get lookup fields & hapus variabel yang tidak digunakan).

Step name: Stream lookup

Lookup step: Microsoft Excel input

The key(s) to look up the value(s):

#	Field	LookupField
1	Nama	Nama
2	Kelas	Kelas

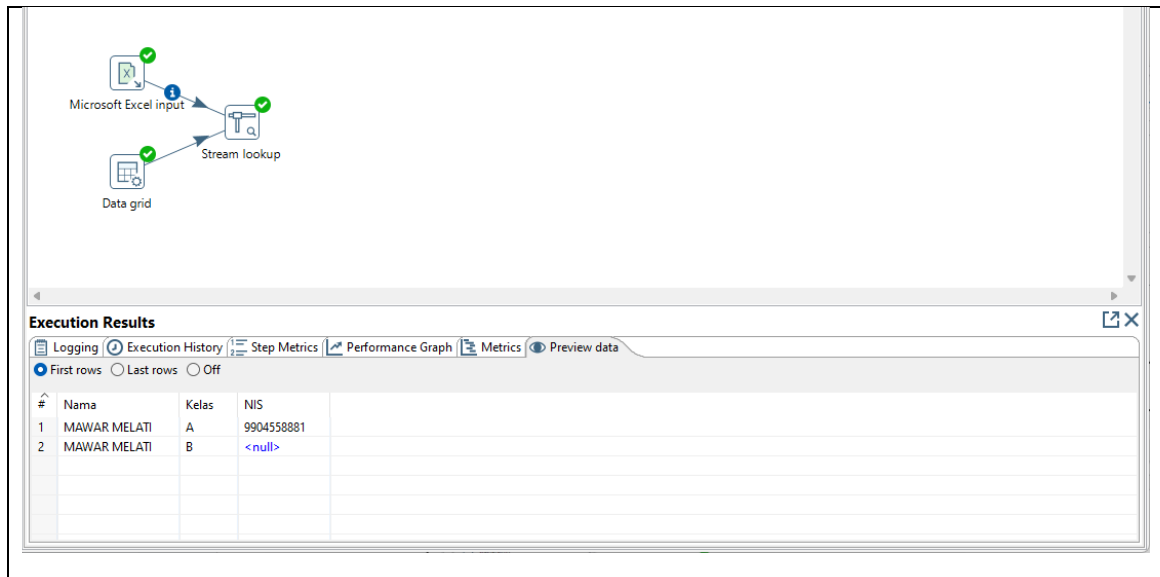
Specify the fields to retrieve :

#	Field	New name	Default	Type
1	NIS			String

☒ Preserve memory (costs CPU)
☐ Key and value are exactly one integer field
☐ Use sorted list (i.s.o. hashtable)

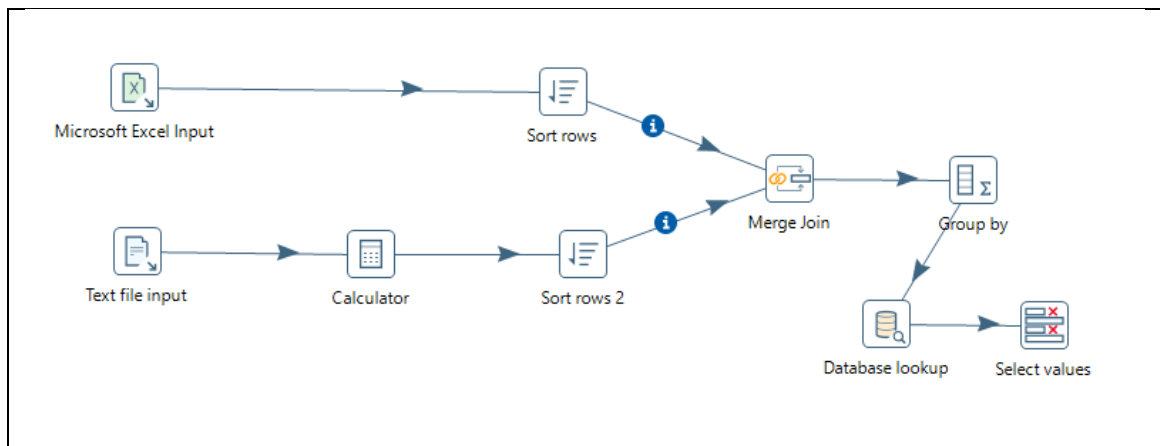


5. Ketika dijalankan maka outputnya akan seperti gambar berikut ini.



b. Latihan Kedua – Penggabungan dengan Merge dan Database Keylookup

1. Nyalakan Apache dan MySQL pada XAMPP dan buat transformasi baru dengan struktur sebagai berikut.



2. Microsoft Excel Input (Get sheetsname dan Get fields).

The image displays three sequential screenshots of the 'Microsoft Excel input' configuration window, illustrating the steps to set up data input from an Excel file.

Top Screenshot: Files Tab
 The 'Files' tab is selected. The 'Step name' is 'Microsoft Excel input'. The 'Spread sheet type (engine)' is 'Excel 97-2003 XLS (XLS)'. The 'File or directory' field is empty, with 'Add' and 'Browse...' buttons. The 'Regular Expression', 'Exclude Regular Expression', and 'Password' fields are also empty. The 'Selected files' table shows one entry:

#	File/Directory	Wildcard (RegExp)	Exclude wildcard	Required
1	C:\Users\bambal\Downloads\Orders.xls			N

 Below the table are checkboxes for 'Accept filenames from previous steps' and 'Accept filenames from previous step', a dropdown for 'Step to read filenames from', and a dropdown for 'Field in the input to use as filename'. A 'Show filename(s)...' button is at the bottom.

Middle Screenshot: Sheets Tab
 The 'Sheets' tab is selected. The 'List of sheets to read' table shows one entry:

#	Sheet name	Start row	Start column
1	Sheet1	0	0

 A 'Get sheetname(s)...' button is at the bottom.

Bottom Screenshot: Fields Tab
 The 'Fields' tab is selected. The 'Additional output fields' table shows the following data:

#	Name	Type	Length	Precision	Trim type	Repeat	Format	Currency	Decimal	Grouping
1	orderNumber	Number	-1	-1	none	N				
2	orderDate	Date	-1	-1	none	N				
3	requiredDate	Date	-1	-1	none	N				
4	shippedDate	Date	-1	-1	none	N				
5	status	String	-1	-1	none	N				
6	customerNumber	Number	-1	-1	none	N				
7	comments	String	-1	-1	none	N				

 A 'Get fields from header row...' button is at the bottom.



3. Text file input (Get fields pada tab fields).

Text file input

Step name: Text file input

File | Content | Error Handling | Filters | Fields | Additional output fields

File or directory: Add Browse...

Regular Expression:

Exclude Regular Expression:

Selected files:

#	File/Directory	Wildcard (RegExp)	Exclude wildcard
1	C:\Users\bamba\Downloads\orderdetails.txt		
2			
3			

Delete Edit

Accept filenames from previous steps

Accept filenames from previous step ☐

Pass through fields from previous step ☐

Step to read filenames from:

Field in the input to use as filename:

Show filename(s)... Show file content Show content from first data line

Help OK Preview rows Cancel

Text file input

Step name: Text file input

File | Content | Error Handling | Filters | Fields | Additional output fields

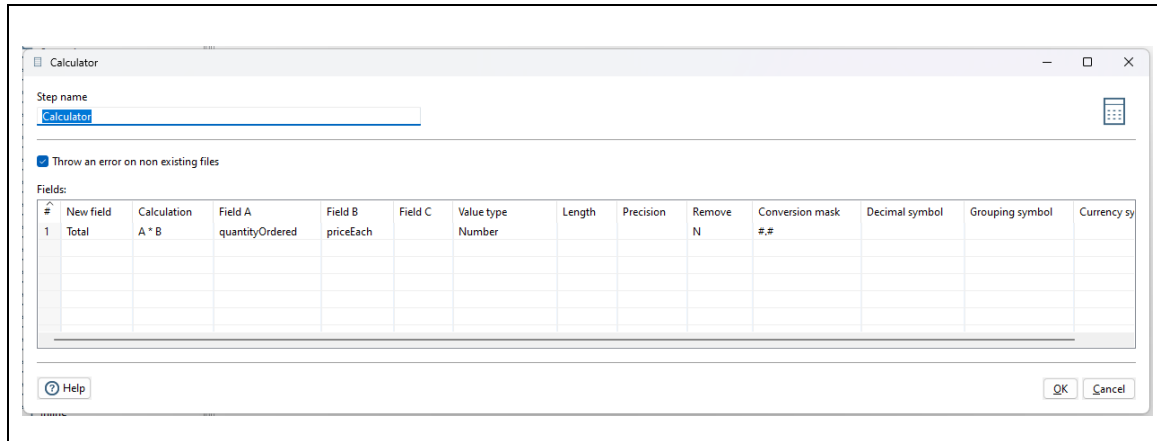
#	Name	Type	Format	Position	Length	Precision	Currency	Decimal	Group	Null if	Default	Trim type	Repeat
1	orderNumber	Integer	#		15	0	IDR	.	,	-		none	N
2	productCode	String			9		IDR	.	,	-		none	N
3	quantityOrdered	Integer	#		15	0	IDR	.	,	-		none	N
4	priceEach	Number	#,.		6	2	IDR	.	,	-		none	N
5	orderLineNumber	Integer	#		15	0	IDR	.	,	-		none	N

Get Fields Minimal width

Help OK Preview rows Cancel



4. Calculator.



Step name:

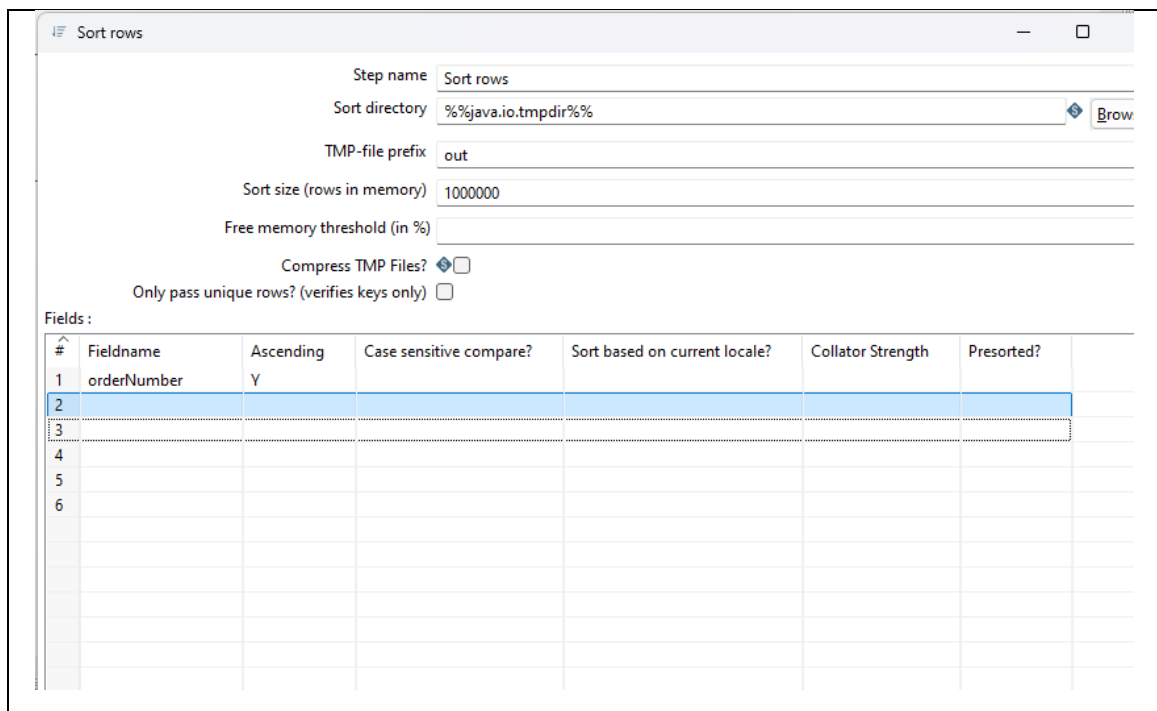
☒ Throw an error on non existing files

Fields:

#	New field	Calculation	Field A	Field B	Field C	Value type	Length	Precision	Remove	Conversion mask	Decimal symbol	Grouping symbol	Currency sy
1	Total	A * B	quantityOrdered	priceEach		Number			N	##			

Buttons:

5. Sort rows (Get fields)



Step name:

Sort directory:

TMP-file prefix:

Sort size (rows in memory):

Free memory threshold (in %):

Compress TMP Files? ☐

Only pass unique rows? (verifies keys only) ☐

Fields:

#	Fieldname	Ascending	Case sensitive compare?	Sort based on current locale?	Collator Strength	Presorted?
1	orderNumber	Y				
2						
3						
4						
5						
6						



6. Sort rows 2 (Get fields 2).

Step name: **Sort rows 2**

Sort directory: %%java.io.tmpdir%%

TMP-file prefix: out

Sort size (rows in memory): 1000000

Free memory threshold (in %):

Compress TMP Files? ☒

Only pass unique rows? (verifies keys only) ☐

Fields:

#	Fieldname	Ascending	Case sensitive compare?	Sort based on current locale?	Collator Strength	Presorted?
1	orderNumber	Y	N	N	2	N

7. Merge join.

Step name: **Merge join**

First Step: Sort rows

Second Step: Sort rows 2

Join Type: INNER

Keys for 1st step:

#	Key field
1	orderNu...

Keys for 2nd step:

#	Key field
1	orderNu...

Get key fields (for 1st step)

Get key fields (for 2nd step)

Help OK Cancel



8. Group by (Get fields & Get lookup fields).

Group by

Step name:

Include all rows? ☐

Temporary files directory: [Browse...](#)

TMP-file prefix:

Add line number, restart in each group ☐

Line number field name:

Always give back a result row ☐

The fields that make up the group:

#	Group field
1	orderNumber

[Get Fields](#)

Aggregates:

#	Name	Subject	Type
1	customerNumber	customerNumber	First value
2	Total	Total	Sum

[Get lookup fields](#)

9. Database lookup (Database connection, Database explorer)

Database Connection

General
Advanced
Options
Pooling
Clustering

Connection name:

Connection type:

- MS Access
- MS SQL Server
- MS SQL Server (Native)
- MariaDB
- MaxDB (SAP DB)
- MonetDB
- MySQL**
- Native Mondrian
- Neoview
- Netezza
- Oracle
- Oracle RDB
- Palo MOLAP Server
- PostgreSQL
- Redshift
- Remedy Action Request System
- SAP ERP System
- SQLite
- Snowflake

Settings

Host Name:

Database Name:

Port Number:

Username:

Password:

☒ Use Result Streaming Cursor

Access:

- Native (JDBC)**
- ODBC
- JNDI

Connection tested successfully

Connection to mysql-classicmodels was successful.

Hostname: localhost
Port: 3306
Database name: classicmodels

OK

Test Feature List Explore

OK Cancel



Database lookup

Step name: Database lookup

Connection: mySQL-classicmodels [Edit...](#) [New...](#) [Wizard...](#)

Lookup schema: classicmodels [Browse...](#)

Lookup table: customers [Browse...](#)

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	customerN...	=	custo...	
2				

Values to return from the lookup table:

#	Field	New name	Default	Type
1	custo...			String

Database lookup

Step name: Database lookup

Connection: mySQL-classicmodels [Edit...](#) [New...](#) [Wizard...](#)

Lookup schema: classicmodels [Browse...](#)

Lookup table: customers [Browse...](#)

Enable cache? ☐

Cache size in rows (0=cache): 0

Load all data from table ☐

The key(s) to look up the value(s):

#	Table field	Comparator	Field1	Field2
1	customerN...	=	custo...	
2				

Values to return from the lookup table:

#	Field	New name	Default	Type
1	custo...			String



8. Select values (Get fields to select, Get fields to change).

Select values

Step name Select values

Select & Alter Remove Meta-data

Fields :

#	Fieldname	Rename to	Length	Precision
1	orderNumber			
2	customerNumber			
3	Total			
4	customerName			

Get fields to select

Edit Mapping

Include unspecified fields, ordered by name ☐

Help OK Cancel

Select values

Step name Select values

Select & Alter Remove Meta-data

Fields to alter the meta-data for :

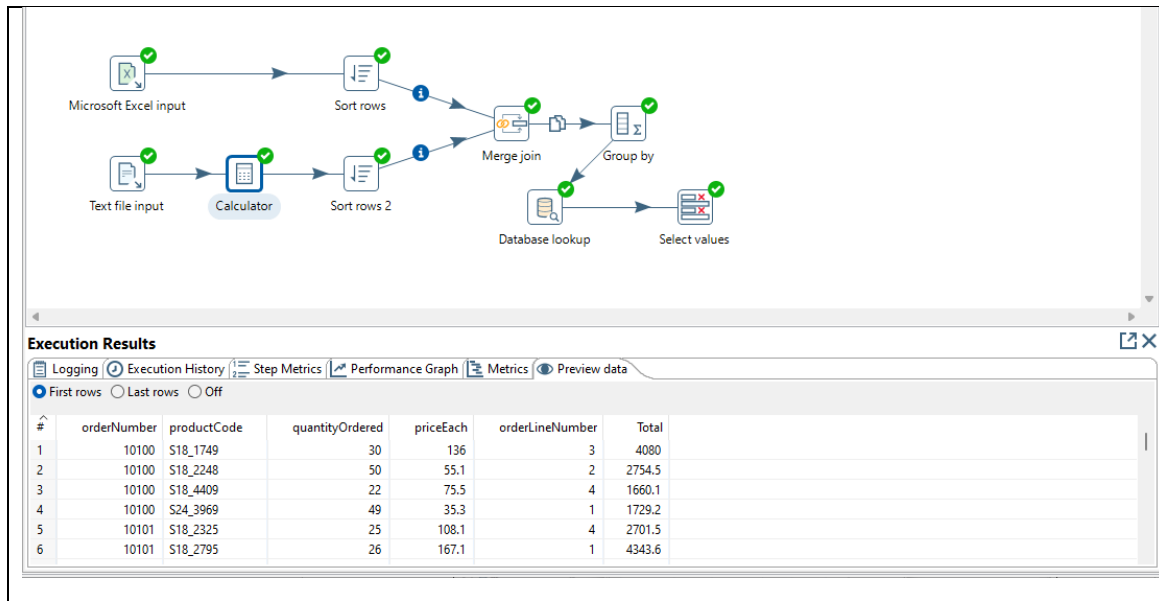
#	Fieldname	Rename to	Type	Length	Precision	Binary to
1	Total		Number			N

Get fields to change

Help OK Cancel



9. Output



4. File Praktikum

Github Repository:

<https://github.com/aryaoctavianus/Praktikum-Datawarehouse/tree/main/Praktikum>

5. Soal Latihan

Soal:

1. Apa yang dimaksud dengan Merge Data?
2. Apa tujuan dan fungsi dari menggabungkan data?

Jawaban:

1. Proses menggabungkan dua atau lebih set data menjadi satu aliran data berdasarkan kriteria tertentu.
2. Untuk menyatukan informasi dari sumber berbeda agar analisis data menjadi lebih lengkap dan menyeluruh.

6. Kesimpulan

- a. Dalam pengerjaan praktikum Data Warehouse, kita harus benar-benar teliti dalam menginputkan suatu fungsi untuk menampilkan suatu keluaran pada layar dengan sesuai.
- b. Kita dapat mengetahui cara menggunakan Merge Data pada Spoon untuk menggabungkan beberapa aliran data menjadi satu output yang terstruktur



7. Cek List (✓)

No	Elemen Kompetensi	Penyelesaian	
		Selesai	Tidak Selesai
1.	Latihan Pertama	✓	
2.	Latihan Kedua	✓	

8. Formulir Umpan Balik

No	Elemen Kompetensi	Waktu Pengerjaan	Kriteria
1.	Latihan Pertama	30 Menit	baik
2.	Latihan Kedua	30 Menit	baik

Keterangan:

1. Menarik ✓
2. Baik
3. Cukup
4. Kurang

